

Evaluating AI in Clinical Practice: A Systematic Landscape Analysis of ClinicalTrials.gov

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Summary

Background

Artificial intelligence (AI) is rapidly reshaping clinical evidence generation, yet the landscape of its clinical validation remains fragmented. Public registries lack the structured metadata required to distinguish AI evaluated as a direct medical intervention (AI comparative clinical trials; e.g., testing a diagnostic algorithm against the standard of care) from AI used merely to facilitate backend research operations (AI-assisted clinical trial tools). We aimed to characterize the functional roles, data modalities, and clinical applications of AI within registered clinical trials.

Methods

We analyzed ClinicalTrials.gov records referencing AI registered through Dec 31, 2025. A validated large language model (GPT-4o) classified trials by functional role, data modality, and clinical application. Investigator-reported conditions were semantically standardized using SapBERT and PheCode mapping.

Findings

Of 2,887 identified trials, 44.7% (n=1,290) were classified as AI comparative clinical trials. These were heavily favored image- and video-based decision-support systems for visual diagnoses (e.g., colon polyps, diabetic retinopathy). Conversely, behavioral applications remained siloed in text and audio, leaving complex systemic diseases underrepresented. Geographically, the national registry skewed the landscape toward the USA and China, underrepresenting European and Japanese trials. Finally, academic institutions sponsored 86.5%

of comparative trials, revealing a structural "transparency gap" regarding commercial AI developers utilizing alternative regulatory pathways.

Interpretation

While clinical AI demonstrates a growing intent toward prospective evaluation, it remains concentrated in single-modality diagnostic silos. Because trial registration does not guarantee methodological rigor, trustworthy deployment requires shifting toward multimodal data fusion for complex diseases. The field must demand decision-grade evidence (prioritizing safety, fairness, and workflow integration) while enforcing AI-specific reporting standards (e.g., SPIRIT-AI, CONSORT-AI, STARD-AI) and auditable architectures to ensure robust global oversight.

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INTRODUCTION

Artificial intelligence (AI) is rapidly reshaping biomedical research and healthcare delivery, achieving state-of-the-art performance across tasks ranging from medical image analysis to predictive modeling¹⁻⁶. However, integrating AI into the rigorous framework of clinical evidence generation poses unique challenges. Beyond the need to process high-dimensional multimodal data, AI interventions require robust, auditable architectures to satisfy evolving regulatory requirements⁷⁻¹¹. The lifecycle of clinical AI typically spans five stages: prepare, develop, validate, deploy, and maintain¹². While much of the early AI literature has heavily focused on the initial preparation and development phases, translating these models into practice requires a massive shift toward the validation and deployment stages. To meet this need, AI's role within the clinical trial ecosystem is rapidly expanding. Its application is shifting from a backend tool for trial execution (e.g., acting as an operational utility for screening patients) to a direct medical intervention requiring rigorous evaluation (e.g., testing an AI diagnostic system directly against a human physician's standard of care)¹³. This transition aligns with growing consensus from clinical informatics leaders and major medical journals, which increasingly advocate that AI interventions must be evaluated using traditional, prospective clinical trial frameworks to compare algorithmic performance against real-world clinical baselines¹⁴.

Even with this surge in activity, a comprehensive understanding of the clinical AI landscape remains limited. Initial landscape analyses primarily characterized AI as an assistive tool for optimizing trial logistics or discovering digital biomarkers^{15,16}, while early reviews of AI

comparative clinical trials were largely restricted to medical imaging¹⁷. More recent narrative reviews have identified emerging data transparency and ethical challenges¹⁸, but largely rely on qualitative synthesis or limited manual cohorts. Furthermore, unstructured metadata in public registries such as ClinicalTrials.gov makes it difficult to distinguish between trials that use AI for backend logistics (AI-assisted clinical trial tools) and those that evaluate it as a direct medical intervention (AI comparative clinical trials)^{19–21}. Consequently, systematic evidence quantifying AI's transition to comparative clinical evaluation remains lacking.

To address these gaps, we conducted a systematic landscape analysis of AI-related clinical trials registered in ClinicalTrials.gov. We developed an automated large language model (LLM) pipeline to classify trial roles, data modalities, and clinical applications. To overcome the heterogeneity of investigator-reported conditions, we applied semantic normalization using SapBERT²² to map conditions to Unified Medical Language System (UMLS) concepts, followed by systematic mapping to PheCodes^{23,24}. Using this framework, this study aims to: (1) quantify the field's transition toward rigorous clinical validation by distinguishing AI-assisted tools from AI comparative trials; (2) map structural patterns by analyzing how specific data modalities channel into distinct functional applications; and (3) expose the dominant disease targets, sponsorship patterns, and transparency gaps currently driving AI clinical research.

METHODOLOGY

Data Source and Search Strategy

We performed a systematic landscape analysis of the ClinicalTrials.gov registry to characterize the evolving role of AI in clinical research. ClinicalTrials.gov was selected as the exclusive data source because its highly structured metadata was strictly necessary for our automated LLM pipeline; expanding to less-structured databases would compromise classification uniformity.

To identify relevant studies, we searched the registry for all records registered through December 31, 2025, using the broad search terms "Artificial Intelligence" and "AI" to maximize sensitivity.

This explicit nomenclature was chosen because it signals deliberate clinical, commercial, or regulatory positioning, distinct from broader computational methodologies. We retrieved all matching records, regardless of recruitment status, study phase, or geographic location, and extracted complete metadata fields for downstream processing. In addition to characterizing the evolving role of AI in clinical research, we established a hierarchical classification framework consisting of three primary dimensions: trial role, data modality, and clinical application.

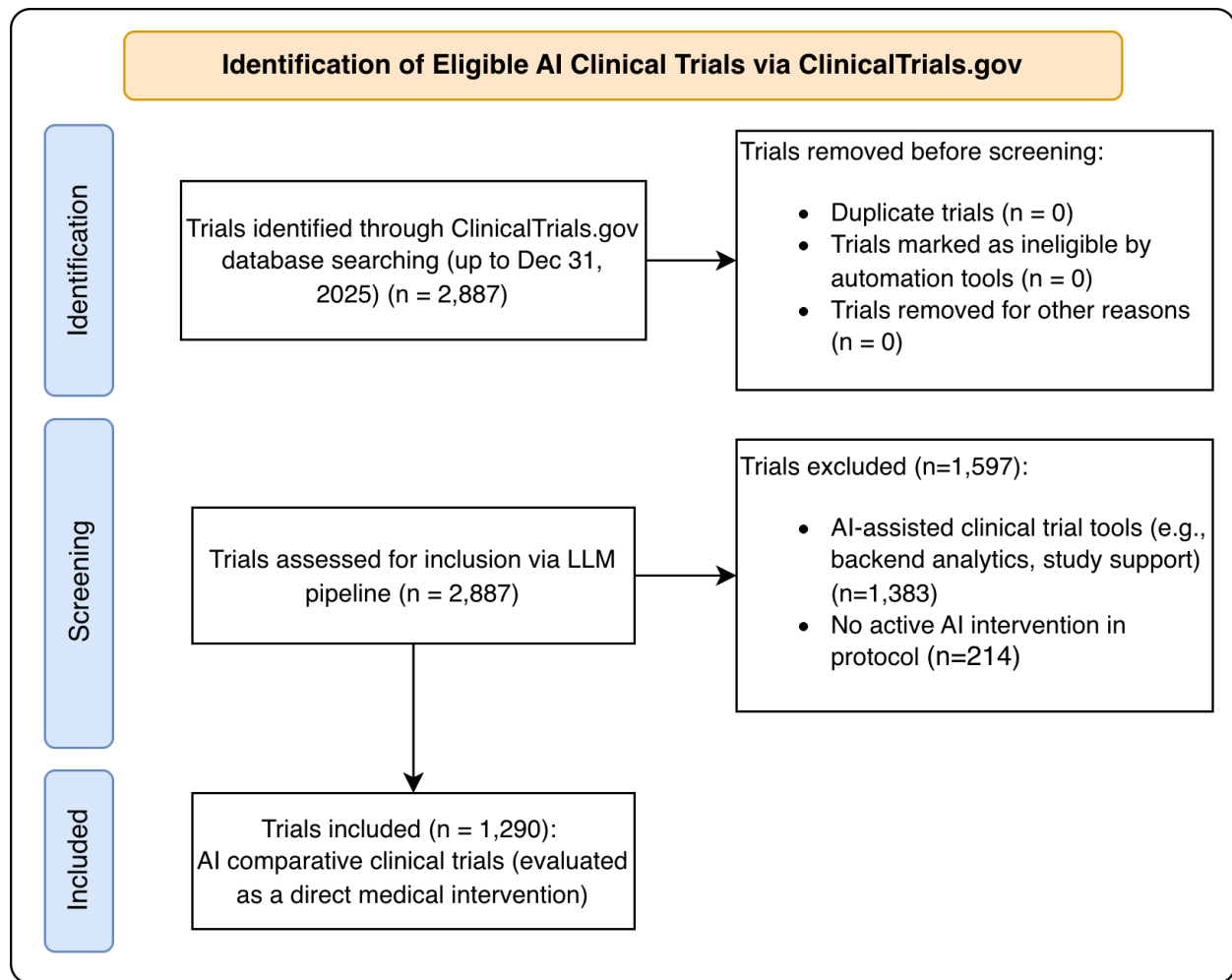


Figure 1. Study Selection Flowchart. The identification, screening, and inclusion of eligible AI clinical trials from ClinicalTrials.gov (registered through December 31, 2025). Of the 2,887 initial records screened, 1,597 were excluded primarily because AI was utilized strictly as a backend operational tool or lacked an active AI intervention in the study protocol. The final analytical cohort consists of 1,290 AI comparative clinical trials evaluating AI as a direct medical intervention.

Classification Framework and Taxonomies

A preliminary manual inspection of the retrieved dataset revealed significant heterogeneity in how AI was incorporated into clinical trials. Many trials referenced AI only peripherally or as a background rationale rather than as an experimental intervention. To separate comparative

evaluations from studies utilizing AI merely as an operational tool, we classified the trials into three mutually exclusive roles based on their primary, registered clinical objective:

- **AI Comparative Clinical Trials:** Studies that evaluate an AI system as the primary medical intervention, directly comparing its clinical performance against a standard-of-care baseline or a non-AI control group.
- **AI-Assisted Clinical Trial Tools:** Studies that use AI as a backend tool to support research operations (e.g., optimizing patient recruitment or analyzing data) rather than testing the AI itself as a medical intervention.
- **Other Role:** Trials that mention AI in the study protocol, but where the technology plays an incidental or undefined role rather than functioning as an active component of the study design.

Because the AI Comparative Clinical Trials cohort represents the primary evidence base for clinical efficacy, subsequent analyses were restricted to this group. For these retained trials, we further characterized the technical and functional landscape using two additional taxonomies:

- **Data Modalities:** Classified into eight categories: Image, Video, Time Series, Sensor Data, Structured Data, Free Text, Audio, and Other Data Modality.
- **Clinical Applications:** Categorized into seven functional groups: Clinical Decision Support Systems (CDSS), Predictive Health Modeling, Telemedicine and Remote Monitoring, Rehabilitation and Assistive Technologies, Behavioral and Mental Health Analytics, Personalized Medicine, and Other Applications.

Automated Annotation Pipeline

Given the scale of the dataset and the complexity of unstructured clinical text, we employed GPT-4o (API-version 2024-06-01 with default inference hyperparameters in zero-shot setting) to automate screening and classification. For each trial, the model was provided with a concatenated text block derived from the Trial Title, Brief Summary, Detailed Description, Intervention, and Outcome fields. We engineered a specific prompt to analyze the semantic context of these fields and perform a two-step annotation: first, assigning one of the three predefined AI roles; and second, identifying all applicable data modalities and clinical applications. The complete text of the prompt and the accompanying classification instructions are provided in Supplementary File S1.

We validated the pipeline's reliability by comparing its output against a gold-standard dataset of 100 stratified trials, independently coded by two experts ($\kappa = 0.85$). The automated system showed strong alignment with human judgment, yielding a $F1 = 0.9637$ for AI Role classification. Reliability remained high for Data Modality ($F1 = 0.9494$) and Clinical Application ($F1 = 0.9461$). These metrics demonstrate that the pipeline is a valid, high-fidelity alternative to manual curation for large-scale datasets.

Semantic Standardization of Medical Conditions

Investigator-reported clinical terms in the "Condition" metadata field often exhibit significant lexical variability (e.g., "Heart Attack" vs. "Myocardial Infarction"). To enable precise

quantitative analysis, we implemented a multi-stage normalization pipeline to map these raw terms to PheCodes. PheCode is a classification system originally developed for Phenome-Wide Association Studies (PheWAS) to aggregate granular diagnostic (ICD) codes into distinct, biologically meaningful clinical phenotypes. By normalizing conditions to this level, we aim to reduce the sparsity and dimensionality inherent in raw clinical terminologies while facilitating more robust statistical comparisons.

First, we normalized raw condition strings to UMLS concepts. We used SapBERT, a biomedical-specific language model, to generate embeddings from raw strings and map them to the nearest UMLS Concept Unique Identifier (CUI), thereby ensuring robustness against synonymy and lexical variation.

Following the initial UMLS normalization, we applied a hierarchical mapping strategy to assign PheCodes:

1. **Primary Mapping:** UMLS CUIs were directly mapped to ICD-10 codes using the concept's source vocabulary identifiers.
2. **Secondary Mapping:** For CUIs without a direct ICD-10 match, we retrieved the associated SNOMED CT Fully Specified Name (FSN) to serve as an intermediate bridge to ICD-10.
3. **PheCode Conversion:** To aggregate diagnostic terms into clinically meaningful phenotypes, we mapped ICD codes using the PheCode version 1.2 combined mapping table.

4. **Retention of Unmapped Concepts:** Concepts failing both mapping pathways were retained as normalized UMLS CUIs to preserve granular clinical information.

Finally, we grouped all identified PheCodes by their root concepts (represented by the integer portion of the code) and, when available, extracted Medical Subject Headings (MeSH) terms directly from trial metadata. For interpretability, we applied specific curation rules: we excluded highly frequent but clinically nonspecific terms (e.g., generic descriptors or healthy cohorts) and simplified complex descriptions into concise clinical labels (Supplementary File S1). We then analyzed the frequency distributions and overlap of these standardized terms to assess consistency between investigator-reported conditions and standardized indexing.

Statistical Analysis and Clinical Contextualization

Descriptive statistics were employed to characterize the distribution of trials across technical, clinical, and administrative domains. To elucidate the relationship between technical design and clinical intent, we performed a multi-dimensional analysis linking data modality, clinical application, and medical conditions.

Specifically, we constructed a descriptive matrix stratifying the dataset by unique modality-application pairings (e.g., Medical Imaging utilized for Clinical Decision Support). Within each of these intersections, we identified the most prevalent PheCode categories. We further visualized these relationships using alluvial flow diagrams (Sankey plots), which quantitatively map the transitions from underlying data modalities to downstream clinical applications. This dual approach enables a granular understanding of how clinical context and

data availability jointly shape the evaluation of AI technologies, helping to identify condition-specific adoption patterns and potential structural biases in the current evidence base.

Temporal trends were analyzed based on AI trial start dates from 2012 to 2025 to identify key inflection points in research activity. Geographic distribution was assessed at the national level to pinpoint research hubs. We also examined funding sources (industry versus academic/other) and trial status to evaluate the economic support structures and longitudinal maturity of the AI clinical trial landscape.

RESULTS

Trial Characteristics and Trends

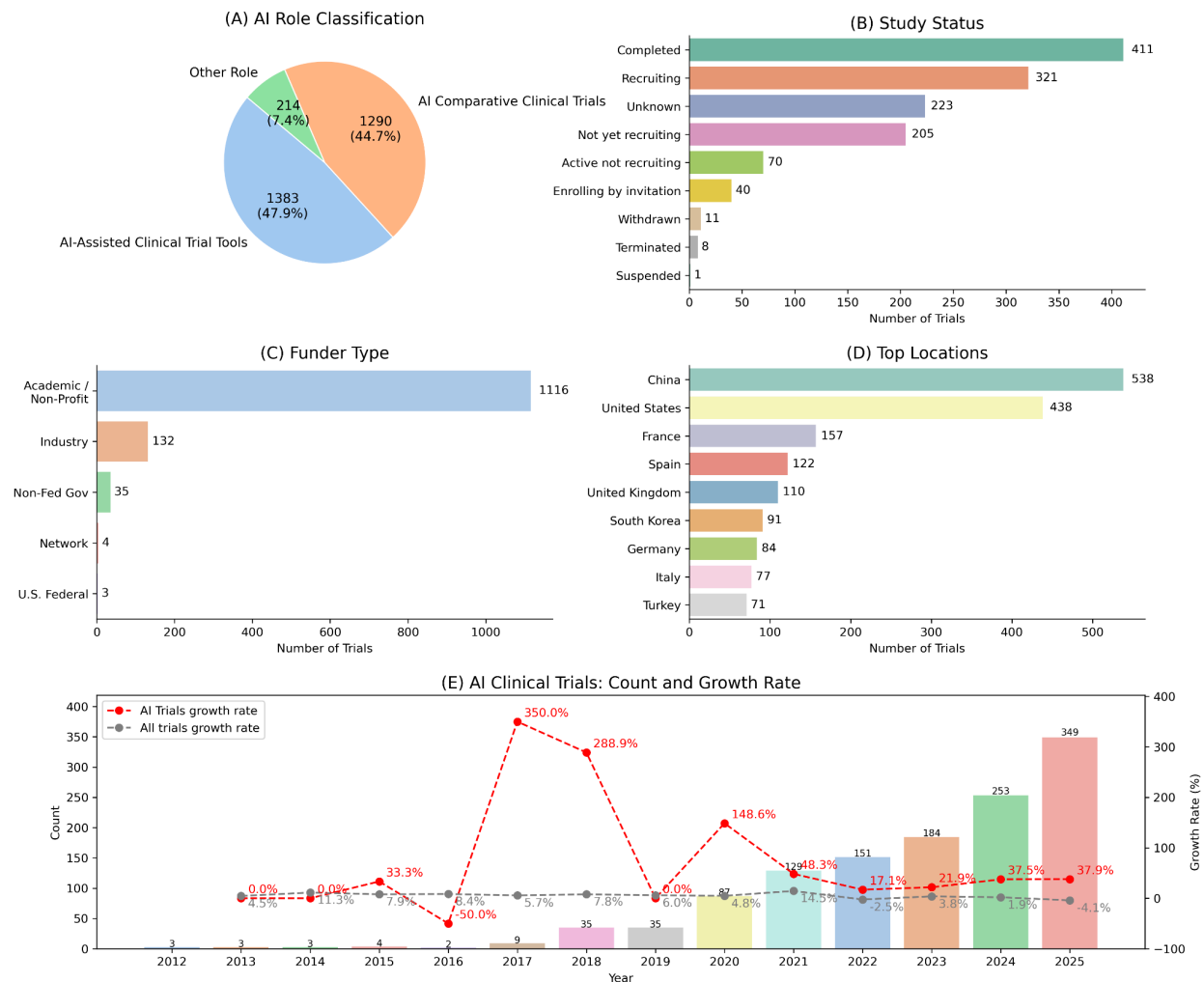


Figure 2. Structural Landscape, Trial Characteristics, and Temporal Evolution of Registered AI Clinical Trials. Summary characteristics of the AI clinical trial landscape. **(A)** The proportion of all 2,887 screened trials categorized by their functional AI role, separating direct comparative trials from assisted tools. **(B)** Current recruitment and completion status of the 1,290 comparative trials. **(C)** Distribution of primary funding types, demonstrating the dominance of academic/non-profit ("Other") sponsorship over industry and government funding. **(D)** Geographic distribution highlighting the top contributing countries to AI comparative clinical research. **(E)** Longitudinal trends displaying the annual registration count and

year-over-year growth rate of AI clinical trials, contrasted with the baseline growth rate of all registered clinical trials, from 2012 to 2025. Growth rate is calculated as the annual percentage change in registration volume compared to the immediately preceding year.

Our search of ClinicalTrials.gov identified 2,887 trials explicitly referencing AI (Figure 1).

Applying our classification framework revealed a balanced field: AI-Assisted Clinical Trial Tools for backend processes accounted for 47.9% (n=1,383), while Comparative Clinical Trials evaluating AI against a standard of care comprised 44.7% (n=1,290) (Figure 2A). A small minority of studies (7.4%, n=214) had an incidental or undetermined role.

Study status indicates that the field is in an active phase of evidence generation (Figure 2B).

Among the comparative trials, 31.9% (n=411) were reported as completed, while 30.3% were actively in progress (recruiting or active but not recruiting). Additionally, 15.9% (n=205) of trials were registered but not yet recruiting, indicating a robust pipeline of upcoming studies. However, reporting consistency remains a challenge: 17.3% (n=223) of trials were classified as "Unknown" status, suggesting potential gaps in long-term registry maintenance.

Analysis of funder types revealed that academic and non-profit institutions ("Other") drove 86.5% (n=1,116) of the comparative trials (Figure 2C). Industry funding played a definitive but secondary role at 10.2% (n=132). Direct government and network funding appeared minimal as the primary classified funder, collectively accounting for only 3.3% (n=42) of the studies.

Geographically, clinical AI research is highly concentrated (Figure 2D). China (n=538) and the United States (n=438) dominate, collectively representing over one-third of registered trials.

Secondary hubs include a strong European block (led by France and Spain) alongside emerging contributions from South Korea and Turkey.

Figure 2E illustrates the growth of AI clinical trials from 2012 to 2025. The bar plot demonstrates an acceleration in AI trial volume, increasing from fewer than 5 annual registrations before 2017 to a peak of 349 in 2025. Comparative trend lines show a distinct divergence after 2020: while the baseline year-over-year growth rate for all registered clinical trials stagnated near zero or even turned negative, the growth rate for AI comparative trials maintained sustained double-digit expansion, reaching 37.9% in 2025..

Technical Implementation and Data Flow

Data modalities were heavily dominated by visual information, specifically image (n=593) and structured data (n=240) (Figure 3A). Video and free text were equally represented (n=199 each), followed by smaller subsets of time series, sensor, and audio data.

Clinical applications overwhelmingly focused on immediate decision support; CDSS comprised 975 trials, exceeding all other categories combined (Figure 3B). Predictive health modeling (n=114) and telemedicine (n=95) formed the next tier, alongside smaller clusters in behavioral analytics, rehabilitation, and personalized medicine.

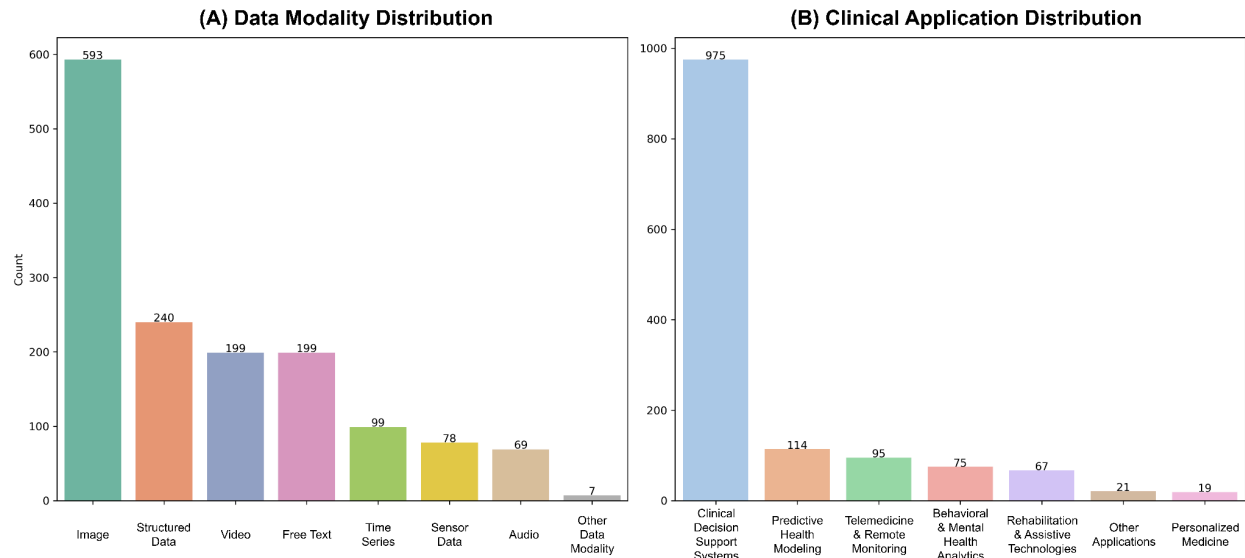


Figure 3. Primary Data Modalities and Clinical Applications of AI Comparative Trials. The distribution of the 1,290 AI comparative clinical trials across predefined technical and functional categories. **(A)** Trial frequency by utilized data modality, highlighting the strong dominance of visual data (Images and Video). **(B)** Trial frequency across broad clinical applications, showing a disproportionate concentration of AI implementations in CDSS.

Alluvial flow analysis revealed distinct data-to-application pipelines (Figure 4). Image and video data predominantly fueled CDSS ($n=554$ and $n=158$, respectively), with video demonstrating secondary utility in rehabilitation ($n=27$). Structured data and time series modalities focused on both CDSS ($n=150$ and $n=76$) and predictive health modeling ($n=69$ and $n=20$). In contrast, sensor data bridged multiple domains, distributed across telemedicine ($n=36$), CDSS ($n=24$), rehabilitation ($n=21$), and predictive modeling ($n=16$). While free text and audio modalities contributed significantly to CDSS, they showed a disproportionate trend toward behavioral and mental health analytics ($n=53$ and $n=12$) compared to other modalities.

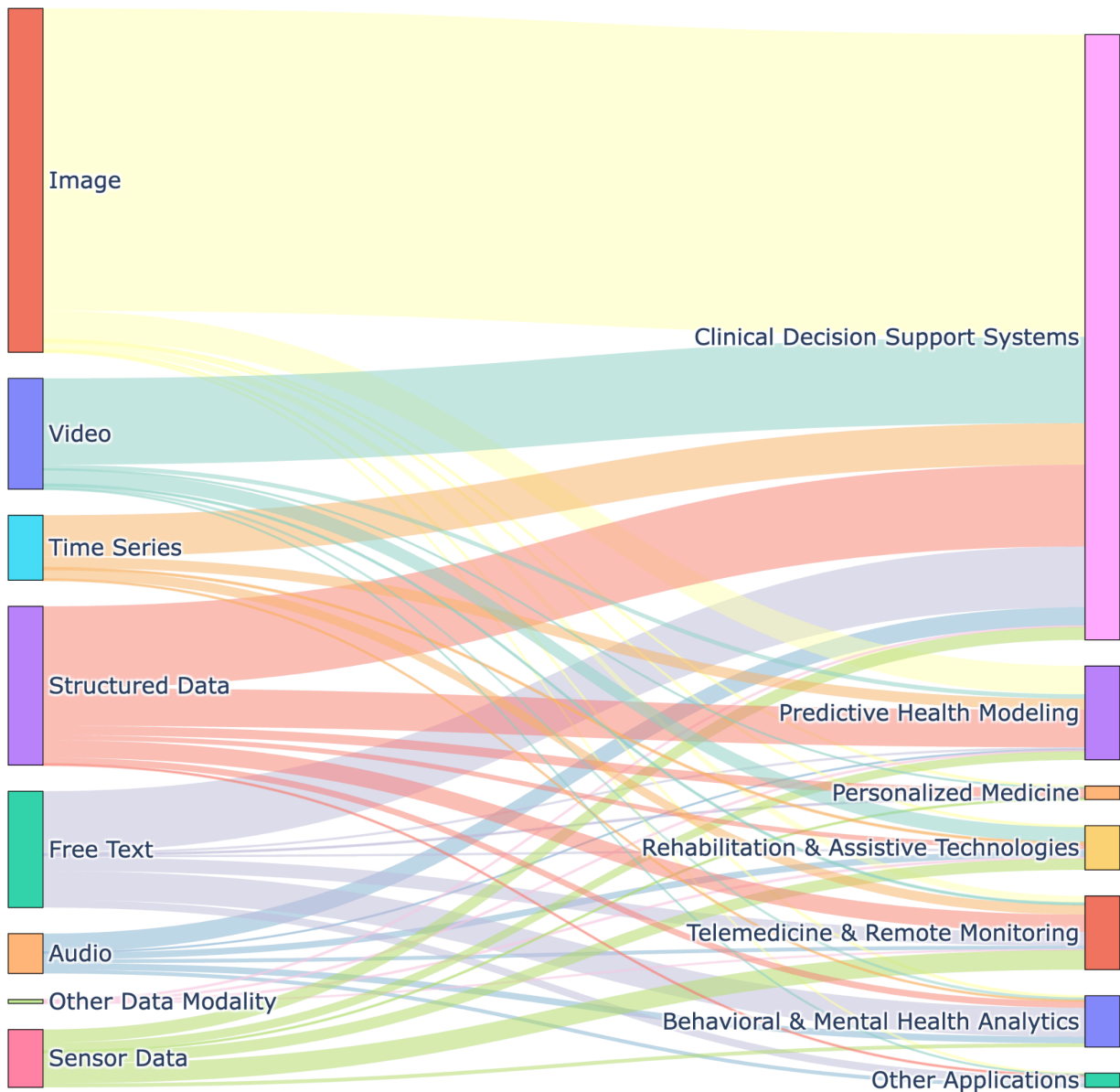


Figure 4. Alluvial Flow Mapping of Data Modalities to Clinical Applications. An alluvial diagram illustrating the structural transition pipelines from technical inputs (Data Modalities, left) to their functional implementation (Clinical Applications, right). Bandwidths are proportional to the number of trials. The visualization highlights dominant developmental pathways, such as the heavy reliance of CDSS on image and video inputs, and the isolated utilization of free text and audio modalities for Behavioral and Mental Health Analytics.

Clinical Context and Disease Focus

While 39.77% (n=1,095) of concepts remained unmapped to PheCodes, successfully mapped conditions (Figure 5A) were led by Colon Polyps/Adenoma (n=105), Diabetes Mellitus (n=92), and Colorectal Carcinoma (n=47). Cardiovascular and oncologic priorities followed, including Coronary Artery Disease (n=43), Breast Cancer (n=43), and Heart Failure (n=40). Extracted MeSH terms (Figure 5B) corroborated these findings, identifying Colorectal Cancer (n=67), Colon Polyps (n=64), and Breast Cancer (n=47) as dominant topics, while also highlighting Diabetic Retinopathy (n=39) and Anxiety Disorders (n=26).

Mapping clinical targets across modality-application intersections revealed distinct diagnostic specializations (Figure 5C). Image-based CDSS (n=554) focused on Diabetes Mellitus and Breast Cancer, while video-based CDSS (n=158) targeted Colon Polyps and Colorectal Carcinoma. Free text and audio data were concentrated in Behavioral & Mental Health Analytics for Anxiety, Mood, and Developmental Disorders. Sensor data bridged telemedicine and monitoring for Diabetes and Heart Failure, while structured data supported CDSS for Sepsis and predictive modeling for cardiovascular disease. Finally, time series data was primarily aligned with cardiac monitoring for Arrhythmia and Heart Failure.

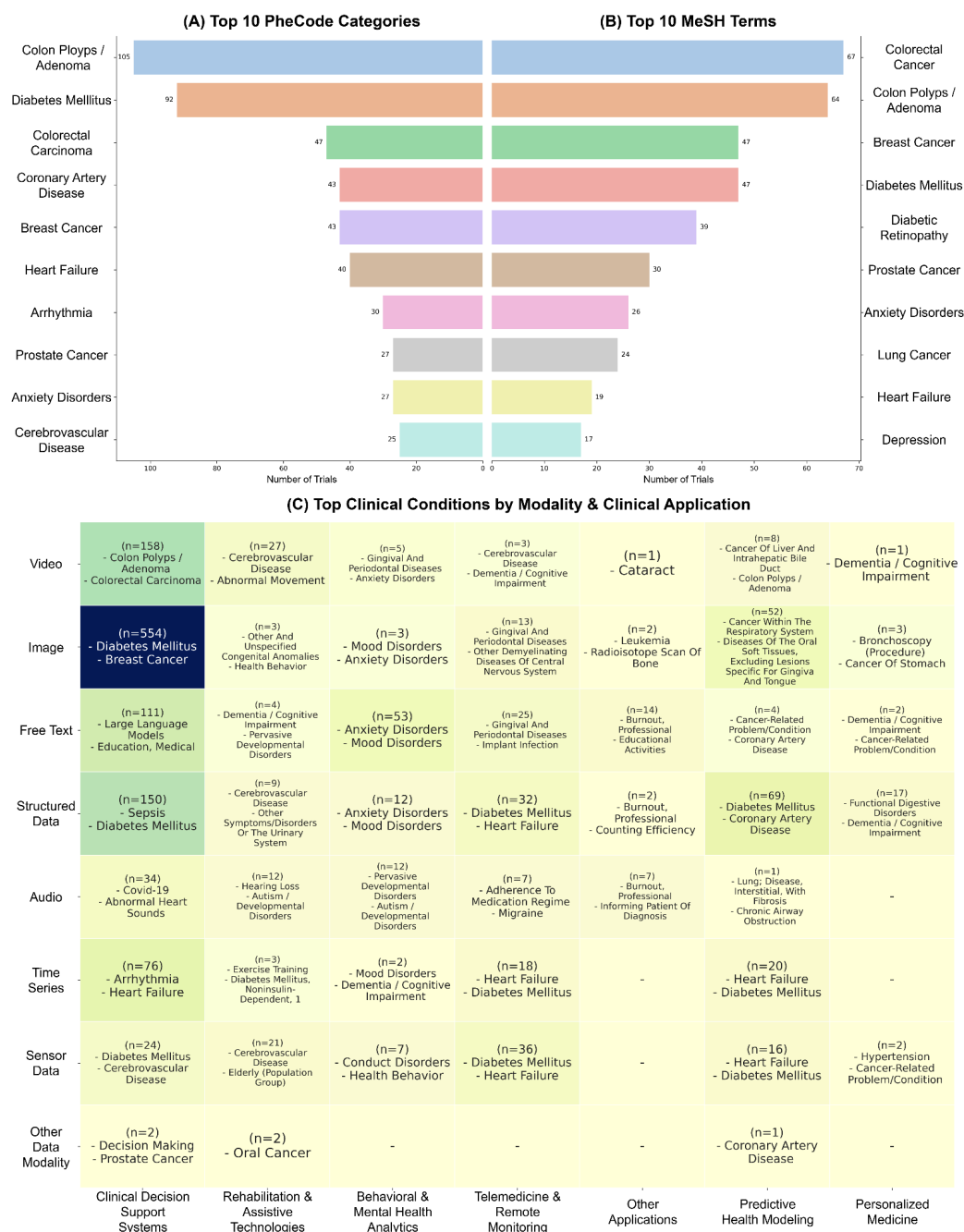


Figure 5. Clinical Context and Dominant Disease Targets in AI Clinical Trials. A multi-dimensional analysis of the specific medical conditions targeted by the AI comparative trials. **(A)** The top 10 most frequently targeted clinical phenotypes, standardized and mapped via PheCodes. **(B)** The top 10 most prevalent MeSH terms extracted directly from trial metadata. **(C)** A cross-tabulation matrix of data modalities and clinical applications, annotated with the most prevalent clinical conditions driving each intersection. The color gradient reflects the total volume of studies within each specific modality-application pairing, demonstrating the siloing of specific disease states into distinct technical pipelines.

DISCUSSION

This study provides a comprehensive landscape analysis of registered AI clinical trials from ClinicalTrials.gov, revealing structural patterns that characterize the current development of clinical AI. Our results demonstrate a striking post-2020 decoupling in research activity. While the overall growth rate for all clinical trials stagnated or contracted—likely reflecting the severe disruption of traditional, site-based clinical research caused by the COVID-19 pandemic—AI comparative trials maintained robust, double-digit expansion. This divergence suggests that the pandemic acted as a dual-force mechanism: while it suppressed traditional trial enrollment and hospital-based research, it simultaneously catalyzed massive investments in digital health. The forced global pivot toward remote monitoring, decentralized trial infrastructures, and computational medicine provided a fertile, insulated environment for AI development, fueling its rapid expansion across the clinical spectrum.

Nearly half of the registered studies focused on AI-assisted tools for operational tasks, such as patient recruitment, cohort identification, and trial monitoring. Concurrently, a comparable proportion aimed to evaluate AI systems as direct medical interventions. This dual deployment reflects a growing research intent to integrate AI into both the conduct of clinical trials and the practice of medicine. While registration is only the initial step in the evidence pipeline, these trends indicate a significant ambition to modernize clinical workflows. However, trial registration must not be conflated with clinical impact. The volume of registered trials provides

no guarantee of methodological rigor, successful patient enrollment, or eventual peer-reviewed publication. Only if these trials are rigorously executed, adequately powered, and successfully published will they have the potential to introduce novel decision-support tools into patient care.

Our analysis of data modalities further highlights the central role of visual data in current clinical AI research. Imaging and video-based trials dominated the landscape, particularly in applications involving CDSS. This pattern aligns with the rapid clinical translation of the deep learning revolution that followed the introduction of foundational CNN architectures (such as ResNet²⁵ and U-Net²⁶), which enabled reliable analysis of medical images at near-clinician-level performance. Our results suggest that these technical advances have translated into a highly specific clinical focus: the prominence of diabetic retinopathy and colon polyps as leading targets reflects a pragmatic strategy to automate high-volume, visually intensive diagnostic tasks.

In contrast to imaging modalities, sensor-derived data demonstrated the most diverse application profile across clinical tasks. Rather than concentrating on a single clinical application, sensor data spanned telemedicine, predictive modeling, rehabilitation, and decision-support systems. This versatility likely reflects the growing adoption of wearable technologies and remote monitoring platforms, which enable the continuous measurement of physiological signals in real-world settings. As healthcare increasingly shifts toward longitudinal and home-based care models, sensor-based AI systems will likely play an expanding role in chronic disease monitoring and the early detection of clinical deterioration. However, facilitating large-scale growth in the wearable AI space will require further technological advancements, particularly in

native device processing (e.g., edge computing), to efficiently handle high-bandwidth physiological data without relying exclusively on continuous cloud connectivity.

The current focus on specific data modalities and clinical applications has isolated the conditions targeted by AI into distinct diagnostic silos. Conditions defined by structural biomarkers (particularly in ophthalmology, gastroenterology, and oncology) are advancing rapidly via the visual pipeline. Conversely, psychiatric and neurodevelopmental conditions remain segregated within a "digital phenotyping" niche dominated by text and audio data. This strict division between somatic and behavioral applications leaves complex, systemic diseases underrepresented. Because complex chronic disease management often spans these domains, clinical AI must shift toward multimodal data fusion. For instance, capturing the full disease burden of heart failure requires integrating physiological sensors with speech patterns to identify concurrent depression. Building the computational infrastructure for this multimodal fusion will necessitate strategic partnerships between academic medical centers and the private technology sector.

Although academic institutions sponsor 86.5% of comparative AI trials, this figure likely underestimates private-sector involvement due to reporting biases. Commercial developers frequently utilize regulatory pathways (e.g., FDA 510(k) or De Novo) that do not mandate prospective public trial registration. Furthermore, industry sponsors often outsource AI analytics or embed unlabeled proprietary algorithms within broader clinical protocols, obscuring their presence in standard metadata. However, genuine adoption barriers also contribute to this low

commercial rate, particularly infrastructural limitations and a lack of robust evidence proving AI definitively reduces trial timelines or costs.

The geographic distribution of trials highlights the increasingly regional nature of clinical AI development, with China and the United States emerging as the two largest contributors. While this pattern broadly reflects strong research ecosystems, data availability, and national investments in digital health, it must be interpreted within the context of global registry dynamics. For example, China's prominent representation (n=538, the highest globally) is likely amplified by National Medical Products Administration (NMPA) guidelines, which often require ClinicalTrials.gov registration for drug and device trials. Conversely, trials originating in Europe and Japan are systematically underrepresented. Regional mandates dictate primary registration on platforms like EudraCT/CTIS and jRCT, limiting consistent cross-registration on US platforms.

As clinical AI matures and moves toward broader implementation, improving adherence to established reporting standards, such as the SPIRIT-AI, CONSORT-AI, DECIDE-AI, and STARD-AI²⁷⁻³¹, will be important to ensure transparency and enable the research community to accurately track the development and validation of AI-based clinical tools. Furthermore, aligning with emerging editorial standards across the medical literature, mandating the preregistration of AI interventions is a critical next step to mitigate reporting biases and monitor the global safety and efficacy of these tools³². Beyond standard reporting, as AI systems increasingly utilize adaptive algorithms, developing robust mechanisms for algorithmic auditability will be required to maintain regulatory compliance. Ultimately, advancing care for complex systemic diseases

will require a deliberate shift toward multimodal data fusion. As this ecosystem expands, demanding decision-grade evidence on fairness and workflow integration—rather than discriminative performance alone—and enforcing transparent registry reporting will be critical to bridging the gap between algorithm development and trustworthy clinical deployment.

This study has four primary limitations. First, restricting our search to "Artificial Intelligence" and "AI" omits broader technical terms, though this intentionally isolates trials with deliberate clinical, commercial, or regulatory positioning from general computational research. Second, as discussed, relying solely on ClinicalTrials.gov introduces structural geographic biases that underrepresent regions with independent national registry mandates. Third, this landscape analysis evaluates research intent rather than executed evidence. We did not extract study design parameters (e.g., sample sizes, blinding protocols) or track completion and publication rates. Therefore, the high volume of comparative trials cannot be directly equated with methodological rigor. Finally, although nearly 40% of clinical concepts failed PheCode mapping, these largely represent non-specific descriptors or healthy cohorts, preserving the integrity of our primary disease targets. Despite these constraints, our systematic approach provides a robust, standardized overview of the AI clinical trial landscape.

To build upon this landscape, future research must assess the downstream clinical, regulatory, and operational outcomes of these AI implementations. Analyses should track completed AI comparative trials to determine their rates of regulatory clearance and real-world deployment. To evaluate operational efficiency, studies must compare metrics such as time to full recruitment between AI-assisted and propensity-matched, non-AI trials. Finally, prospective, randomized

trials are needed to evaluate the real-world efficacy of multimodal AI systems for complex diseases against standard-of-care baselines.

Research in context

Evidence before this study

We searched PubMed and Google Scholar for systematic reviews and landscape analyses of artificial intelligence in clinical trials published between Jan 1, 2018, and Dec 31, 2025. Search terms included "artificial intelligence", "machine learning", "clinical trials", and "landscape analysis". Previous literature has often struggled to distinguish between the distinct functional roles of AI in clinical research. Initial reviews (e.g., Harrer et al.¹⁵ and Shah et al.¹⁶) primarily characterized AI as an operational tool for optimizing trial logistics, such as patient recruitment and stratification, or as a mechanism for discovering digital biomarkers, rather than as a standalone clinical intervention. While some other studies did examine AI as a direct clinical intervention (e.g., Mayorga-Ruiz et al.¹⁷), these analyses were explicitly restricted to medical imaging, establishing a historical focus on radiomics and visual data. More recent narrative reviews (e.g., Askin et al.¹⁸) have correctly identified emerging challenges in data transparency and ethics, but these largely relied on qualitative synthesis or limited manual cohorts. Consequently, there is a lack of systematic, quantitative evidence measuring the extent to which the field has shifted from using AI merely as a backend operational utility to conducting rigorous

comparative clinical trials that evaluate AI systems as direct medical interventions against traditional standard-of-care baselines.

Added value of this study

To our knowledge, this is the first study to employ an automated large language model and semantic normalization pipeline (PheCode mapping) to systematically classify the functional roles of AI across 2,887 registered trials from clinicaltrials.gov. Our analysis provides four novel contributions. First, we demonstrate that nearly half (44.7%) of all AI-related activity reflects a registered ambition to conduct comparative evaluation against standard-of-care baselines, signaling a desire to move beyond retrospective piloting. Second, we identify a distinct concentration in visual data: imaging and video data overwhelmingly dominate clinical decision support systems, while behavioral applications remain siloed within unstructured text and audio modalities. Third, we reveal a structural transparency gap; although 86.5% of AI comparative trials list academic sponsors, this likely underrepresents industry involvement due to outsourced analytics and the embedding of proprietary algorithms within broader protocols. Finally, we map the global footprint of AI from clinicaltrials.gov, demonstrating how regional registry mandates artificially skew the visible landscape toward the US and China while systematically underrepresenting European and Japanese trials.

Implications of all the available evidence

AI research is shifting from exploratory operational tools toward prospective clinical evidence generation. However, its clinical application remains distinctly unbalanced. Current research heavily favors automating diagnostic tasks for conditions with observable visual biomarkers (such as colon polyps or diabetic retinopathy) while lagging in applying multimodal AI to complex, systemic diseases. To ensure broader clinical relevance, future trials must integrate diverse, multimodal data that accurately represent real-world disease management. Furthermore, alternative regulatory pathways and structural reporting gaps in public registries constitute a critical barrier to oversight. Stricter adherence to AI-specific reporting guidelines (including SPIRIT-AI, CONSORT-AI, DECIDE-AI, and STARD-AI^{27–31}) and the development of auditable architectures are essential to accurately track the global safety, efficacy, and private-sector development of these technologies.

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